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Probabilistic Mapping with Ultrasonic Distance Sensors

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Abstract

This paper proposes a probabilistic robotic mapping approach to merge ultrasonic distance readings by modelling them as Gaussian random variables and using scan matching to reduce uncertainty in mapping process. To account for the high angular uncertainty of ultrasonic distance sensors, both positive readings (detected objects) and negative readings (the lack of detection) are taken into account to update the measurements and create the updated environment map. To support this approach, the map consists simultaneously from two parts – free space scans are stored in occupancy grid and obstacle readings are represented as Gaussian variable feature set.

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Keywords: Robotic mapping; Probabilistic mapping; Ultrasonic distance sensors

1. Introduction

One of the fundamental problems in mobile robotics is the environment mapping problem. Robots need to be able to construct a map of the environment and to use it for the navigation and localization. To reduce the computational complexity, it is common to introduce geometric constraints (for example, existence of lines) in mapping[1,2], but it makes such process inapplicable to less structured outdoor environments, which are often featured in agriculture (for example, orchards). To confront this problem, a search for correspondences for raw sensor readings can be used without making assumptions about the structure of environment.

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Ultrasonic distance sensors are less accurate than laser sensors and are therefore less commonly used for mapping in robotic systems. However, these sensors cost significantly less, are lighter in terms of weight and consume less power, which make them an appealing alternative, if the resulting map is sufficiently accurate for robot purposes. Additionally, there are some environments (for example, under water), in which ultrasonic sensors achieve better results than laser sensors[3].

The proposed mapping approach offers a way to merge ultrasonic distance readings by modelling them as Gaussian random variables and using scan matching to reduce uncertainty in mapping process. Scan matching is a well-known concept in robotic mapping, and there are multitude of approaches that use scan matching to create the map of environment[4,5]. However, their common characteristic is that only positive readings (detected objects) are used to update the object locations. The proposed mapping approach uses both positive readings and negative readings (the lack of detection) to update the measurements and create the updated environment map.

2. Probabilistic map model

2.1. Sensor scan representation

Each scan S_k is a list of k distance readings, which are performed from the same robot position in different directions. Scan configurations consisting of several measurements are especially important in the case of ultrasonic distance sensors due to their high uncertainty. The correlations between individual readings in the full scan S_k can be used to significantly reduce the uncertainty caused by the width of the sonar beam. For example, as seen in Fig. 1, three readings R1, R2 and R3, which partly overlap, have been acquired, and the obstacles are detected in R1 and R3, but not in R2. By using the negative information that R2 did not detect an obstacle in the beam's path, the angular uncertainty of R1 and R3 readings can be reduced.

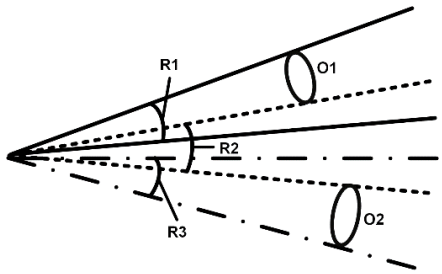

Fig. 1. The reduction of uncertainty of sensor readings by using scan set.

Individual distance readings in the scan S_k are represented as $z_n = [d_n,\ \varphi_n]$, where d_n represents the distance to the closest detected obstacle and the φ_n represents the angle, at which the reading was performed. The robot position $r_t = [x_t, y_t, \theta_t]$ at the moment of scan is also known.

Each reading z_n can be represented as a Gaussian variable by using r_t as a reference point. To achieve that, distance reading Gaussian representation $\tilde{z}_n = [\mu_n, \Sigma_n]$ is calculated for z_n, where μ is Gaussian mean of the detected obstacle and Σ is covariance matrix (see Fig. 2).

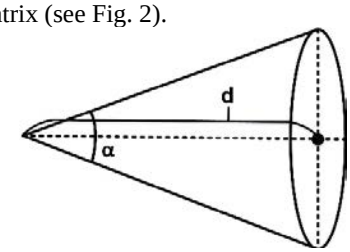

Fig. 2. Distance reading $z_n = [d_n,\ \varphi_n]$ and its Gaussian representation.

The uncertainty of the reading is represented in its covariance matrix Σ (1), which is calculating by using measured distance and width of sonar beam[6]. σ_{dd} represents the distance uncertainty of the reading (2) and $\sigma_{\varphi\varphi}$ represents angular uncertainty of the reading (3), where α is the width of sonar beam:

$$\Sigma' = \begin{bmatrix} \sigma_{dd} & 0 \\ 0 & \sigma_{\varphi\varphi} \end{bmatrix} \tag{1}$$

$$\sigma_{dd} = \frac{d}{60} + 1 \tag{2}$$

$$\sigma_{\varphi\varphi} = \frac{d}{2} tan \frac{\alpha}{2} \tag{3}$$

As seen in Fig. 2, the covariance matrix calculated with (1) represents probability distribution in the reference frame aligned with robot 'zero' angle axis of the robot. The actual covariance of the reading is acquired by transformation as seen in (4):

$$\Sigma = F\,\Sigma' F^T,\ where\ F = \begin{bmatrix} cos\,\varphi & -sin\varphi \\ sin\varphi & cos\varphi \end{bmatrix} \tag{4}$$

All the distance readings in the scan form a set of point features FP_t, where each feature is represented as a Gaussian variable and detected at time t (see Fig. 3a). However, detected obstacles are only a part of the information acquired from the scan. Another part is the free space scan, which is represented as an occupancy grid FG_t (Fig. 3b) and contains the free areas, where the obstacles were not detected. Maximum range sensor readings are not included in the free space occupancy grid. The scan result is a combination of a feature set FP_t, which represents the locations and uncertainty of the environment obstacles, and an occupancy grid, which represents the free area scans. Both of these parts are merged together into the final version of the scan (see Fig. 3c), which is then incorporated in the global map.

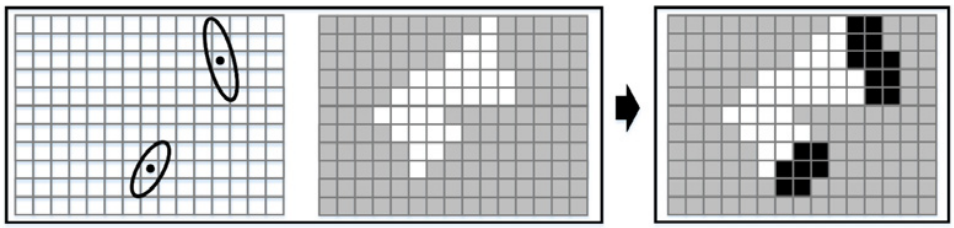

Fig. 3. (a) Feature set; (b) Free space scan; (c) Merging of feature set and free space scan as occupancy grid.

2.2. Feature update

Each reading $\tilde{z}_n$ is matched with the closest existing feature $\tilde{z}_m$ in the map, and if both are considered to be close enough (their Euclidian distance is lower than a preset threshold), the two Gaussian variables are merged into one and the new mean and covariance of the merging is calculated. The resulting variable will have a lower variance than both original ones (see Fig. 4).

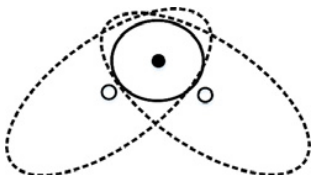

Fig. 4. Illustration of the feature mean and covariance update (dotted lines represent initial features, the continuous line is the result).

The new mean and covariance of the new feature can easily be calculated by applying Kalman filter to both variables. The map feature $\tilde{z}_m = [\mu_m, \Sigma_m]$ serves as the prediction, while the scan $\tilde{z}_n = [\mu_n, \Sigma_n]$ fulfills the role of measurement. Innovation of Kalman filter is calculated as a difference between the mean values of both variables (5). There is no need for transformation matrix H as both variables are already represented in the same reference frame:

$$Y = \mu_n - \mu_m \tag{5}$$

The Kalman gain is calculated from both covariance matrixes (6):

$$K = \Sigma_m(\Sigma_m + \Sigma_n)^{-1} \tag{6}$$

The new mean μ_g and new covariance Σ_g is then calculated as seen in (7-8):

$$\mu_g = \mu_m + KY \tag{7}$$

$$\Sigma_g = (I - K)\Sigma_m \tag{8}$$

To note the fact that the point has been observed more than once, the variable *measuredTimes* is increased. The point of this variable is that the point should be harder to erase, if it has been observed more than once.

2.3. Feature correction with free space scan

Most mapping approaches, which represent obstacles as features, do not inherently use the negative information about free spaces. Some examples of such methods are EKF SLAM[7,8] or particle filter based SLAM[9,10]. However, it can be argued that the lack of obstacle detection is a significant source of information that should not be ignored. To incorporate free space information, the proposed mapping approach represents the free space scan as occupancy grid and then uses it to modify the existing Gaussian variable feature set.

Free space scan information is harder to incorporate in feature $\tilde{z}_m$, because the representations are fundamentally different: free space scans are represented in the form of occupancy grid map, while the obstacles are represented as Gaussian variables. To address this difference of representations, particle sets are used. For each feature $\tilde{z}_m$ in the map, which could be affected by free space scan, a set of particles P_o is randomly generated according to the mean μ_m and the covariance matrix Σ_m and approximated to correspond to individual occupancy grid cells (see Fig. 5a). Each particle in the set P_o represents a hypothesis of obstacle $\tilde{z}_m$ location: $p_k = [x_k, y_k]$.

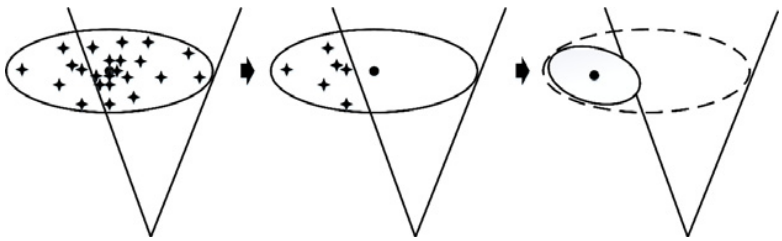

Fig. 5. (a) Free space scan and feature represented as particle set; (b) Some particles are deleted by free space scan; (c) Resulting Gaussian feature.

As a result, the resulting particle set still represents the original Gaussian variable $\tilde{z}_m$, but the individual particles can now be compared to the free space scan in the form of occupancy grid, where each free cell can also be represented as a point $p_l = [x_l, y_l]$. All the P_o particles p_k, which intersect with the set of free points P_f, are deleted

(see Fig. 5b). If there are at least 5% (experimentally set threshold) of particles p_k left in set P_o, then a new Gaussian variable is calculated from the remaining particle set P_o' as seen in (9-10) (see Fig. 5c).

$$\mu = \begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix} \tag{9}$$

$$\Sigma = \begin{bmatrix} \frac{\sum_{i=1}^{k}(x_i-\bar{x})(x_i-\bar{x})}{k-1} & \frac{\sum_{i=1}^{k}(x_i-\bar{x})(y_i-\bar{y})}{k-1} \\ \frac{\sum_{i=1}^{k}(y_i-\bar{y})(x_i-\bar{x})}{k-1} & \frac{\sum_{i=1}^{k}(y_i-\bar{y})(y_i-\bar{y})}{k-1} \end{bmatrix} \tag{10}$$

If more than 95% of the particles in the P_o are deleted by free space scan, then the feature $\tilde{z}_m$ is considered for deletion. However, the $\tilde{z}_m$ is deleted only if its measurement count variable *measuredTimes* is lower than the preset threshold. If it is higher, then the feature is not deleted, but its measurement count variable *measuredTimes* is lowered by one to represent the free space scan observation.

2.4. Scan merging with the map

After free space scan and obstacle readings are incorporated in the feature set, occupied and free parts are merged into common representation – an occupancy grid map. This is a three step process:

- First, the free space scan FG_t is merged with the previous map by using Binary Bayes filter without including the feature set. The result is stored for use in next sensor scan incorporation (see Fig. 6)

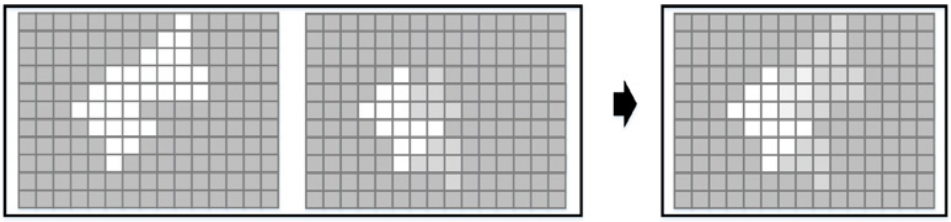

Fig. 6. Merging of free space scan FG_t with previous map without occupied part and the result of merging.

- Next, the feature set FS_t is transformed into occupancy grid, which contains only occupied part of the map. The features are represented as approximated 2σ bound ellipses (see Fig. 7)

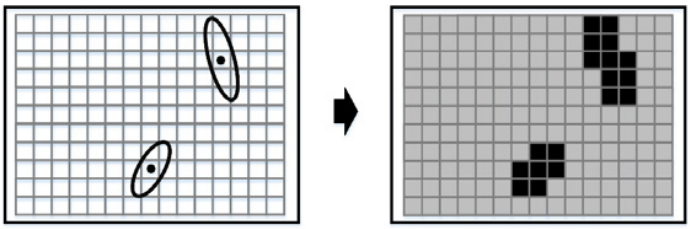

Fig. 7. Feature set FS_t and its approximated representation as occupancy grid.

- Finally, the occupied part map is merged with the free space map to form the final map, which can then be used for robot path planning (see Fig. 8).

The occupancy values of the map cells are updated by using binary Bayes filter[11] which is applied to estimate the binary state of the cells over time. It allows maintaining of a 'memory' of the previous sensor measurements by maintaining additional log odds variable l_t and at the same time ensuring that the value of the cell always assumes a value between 0 and 1.

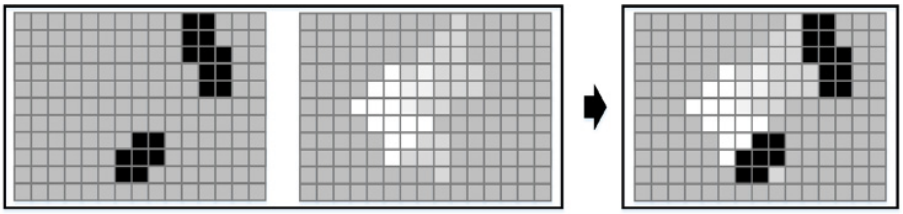

Fig. 8. Merging of feature set represented as occupancy grid and free space map and the result of merging.

The (11-12) show the update of each occupancy grid cell affected by sensor scan:

$$l_t = l_{t-1} + log_{10}(\frac{p(x|z_t)}{1-p(x|z_t)}) - log_{10}(\frac{p(x)}{1-p(x)}) \tag{11}$$

$$p(x) = \begin{Bmatrix} p(x|z_t) & if\ p(x) > 0.5 \\ 1 - \frac{1}{1+\exp(l_t)} & if\ p(x) \leq 0.5 \end{Bmatrix} \tag{12}$$

It (12) shows that the probability of the cell in case of 'occupied' reading (> 0.5) in our case is not actually dependent on the log odds l_t. The reason is that the feature set is already formed taking into account free space measurements and is currently assumed to be correct and is therefore represented in the map.

3. Experimental results

The mapping method described in this paper was implemented in software, which imitates the ultrasonic distance sensor scans. The mapping approach was tested with and without feature correction with free space scan. In Fig. 9a example of environment mapping without the feature correction with free space scan can be seen. Fig. 9b shows environment mapping example, where feature correction with free space scan was performed.

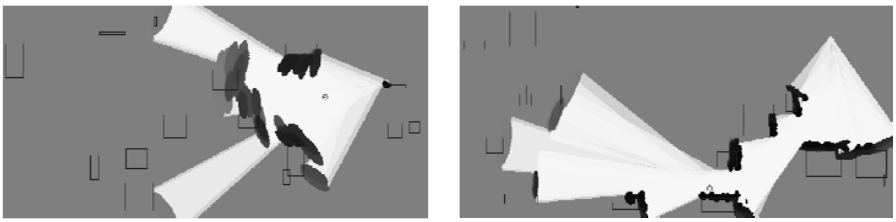

Fig. 9. Map examples (a) without feature correction with free space scan; (b) with feature correction with free space scan.

The testing results show that in the maps where feature correction was performed, the average variance of the features was significantly lower than in the cases where it was not performed. This is not surprising, if we consider that free space scan inclusion in the feature update is an additional information, which is impossible to acquire just from feature scan part.

4. Conclusion

The paper presents a probabilistic mapping method for ultrasonic distance sensors, which incorporates not only positive readings (detected objects), but also negative readings (the lack of detection) into the environment map of the robot to reduce the high uncertainty of sonar readings. Main contribution of the approach is the inclusion of feature correction with free space scans. Although the free and occupied space representations in this case are fundamentally different, the problem is addressed by using particle sets, which act as a bridge between the two representations. The testing results show that the average variance of the features is significantly lower in the cases, when the feature correction with free space scan is performed.

The acquired environment maps acquired are consequently more accurate. In the current version the robot position during the mapping is assumed to be known or with only small deviations. This is currently the main drawback of the approach and limits its application mainly to robotic systems with robust landmark localization, such as the multi-robot system developed in Riga Technical university[12].

To account for the robot position uncertainty, during the next development step the proposed approach will be extended to address simultaneous localization and mapping (SLAM) by applying EKF, which has been widely used before to estimate the robot position in environments that represent the features as Gaussian variables.

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